Faculty of Mathematics and Informatics
Computer Science Department
Animated Multimedia Application Design
Master : Artificial Intelligence & Applications (AIA)
Session 1 Exam –

The standard answer key:
Correction elements

Exercise 1 : Check the correct answer(s): A wrong choice cancels the answer **(06 pts)**

1	The gateways to knowledge are indirect if they	*Use intermediaries.*	
		Result in indirect knowledge	
		Use interfaces to replace physical stimuli	
	0.25 / Correct Choice	*Use inferences and reasoning instead of the 5 direct senses.*	X
2	A sensory channel for acquiring knowledge from the external world defines	*Data input device*	
		Sensorimotor peripheral	
		Perception level	X
		Multimedia application augmentation	
3	Communication between two entities is always	*Complete (Information is transmitted completely and faithfully)*	
		Complete but reformulated (Same content but different expression)	
		Incomplete (Information is never transmitted completely and faithfully)	X
		Incomplete but faithful (missing but the receiver reconstructs the lack)	
4	The auditory dimension in a multimedia scene is	*The simplest dimension to model compared to vision and haptics*	
		The most difficult dimension to implement compared to vision and haptics	
		The simplest dimension to analyze compared to vision and haptics	X
		The most difficult dimension to analyze compared to vision and haptics	
5	The dimension which cannot alone transmit complete cognition to the user is said to be	*Main dimension*	
		Secondary dimension	X
		Autonomous dimension	
		Auxiliary dimension	X
6	The deployment of the visual dimension in a multimedia application is	*Impossible on network if the application is interactive*	
		Possible on a network but expensive if the application is interactive	X
		Possible in network if the rendering is vector only	
		Possible in network if the rendering is matrix only	
7	Visual ordering (the arrangement of visual rendering) in a multimedia application	*Is managed by logical and exact rules only*	
		Is a function of convention (use) only	
		Is a function of the constraints of human perception only	
		Is managed by the three previous constraints	X
8	The mental load (mental charge) induced by the use of a multimedia application	*Increases with the dimension of the application*	
		Increases with the number of channels used by the application	X
		Increases with the number of used devices	
		Increases with quality of rendering devices	
9	In an MMA, a hollow entity	*Is never animated in a scene*	
		Is animated indirectly by modification of other entities only	X
		Is animated indirectly by modifying the values of its pixels	
		Is never static, and is in perpetual motion (all the time)	

10 A frame by frame animation is	*More optimal in terms of resource use than real-time animation*	
	More optimal in rendering quality than real-time animation	
	More optimal in space than real-time animation	
	More optimal in flexibility and design choice (Content Scenario)	X
11 In the 2D production pipeline of classic 2D cartoons	*The production is carried out by the artist and the technician*	X
	The production does not involve the artist	
	The production is carried out by the technician only	
	The production is completely automated	
12 The most important gain brought by CAD in the 2D production chain concerns	*The preproduction step*	
	The production step	
	The post-production	X
	Fairly the three steps	
13 Graphics computing provides to the 2D production line	*Thematization, accuracy, and harmony implemented*	X
	Complete elimination of the human factor from the pre-production phase	
	Complete elimination of the human factor in the mixing and editing phases	
	The reusability of sources (layers and characters)	X
14 The digital palette is	*Richer than the natural palette in number of colors*	
	More exact than the natural palette but less rich	X
	Richer and more exact than the natural palette	
	Less accurate and less rich than the natural palette of colors	
15 Automation of animations is possible in the case where the animation is	*Low number of actors*	
	Rich in number of actors	
	Decomposable into atomic animations	X
	Not decomposable into atomic animations	
16 The expression of emotions on symbolic faces is	*Easier, requires exaggeration, but is less realistic*	X
	Easier, requires exaggeration, and is more realistic	
	Easier, does not require exaggeration, but is less realistic	
	Easier, does not require exaggeration, and is more realistic	
17 Software integration of a multimedia application requires	*Modularity of content and bidirectional management of communication flows*	
	Content singularity and support for interactivity	
	Content modularity and support for interactivity	X
	Singularity of content and bidirectional management of communication flows	
18 FACS is a system allowing	*Automating the rendering and interpretation of facial expressions*	X
	Support for unconventional facial expressions	X
	Decision making when faced with specific facial expressions	
	The automation of human decision-making in the face of virtual faces	
19 Mixed emotions are emotions	*Requiring other emotions to be encoded and expressed*	
	Expressed based on universal emotions	X
	Can be assimilated without facial expressions	
	Easy to formulate and code on their own	
20 The scene is :	*The architectural decomposition of a multimedia application*	
	The atomic temporal unit in a multimedia application	X
	The functional decomposition of a multimedia application	
	All of the three answers above.	

Exercise 2 : (14 pts)

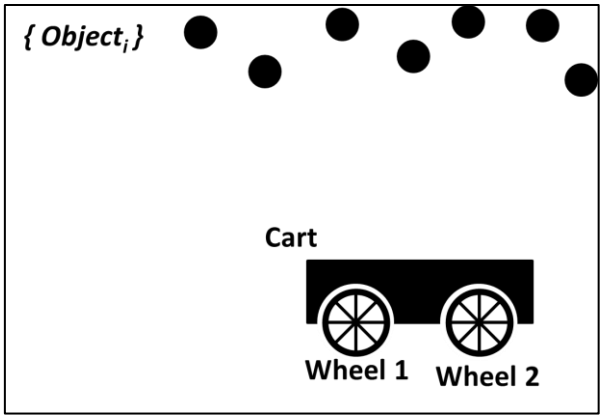
Consider a multimedia application to assess cognition through a simple game:

- The user manages a cart on the horizontal axis by moving it from left to right and vice versa;
- Simple objects will fall and the user will have to catch them using the cart.
- Each time an object is captured, the scene is illuminated (A light is displayed in the background).
- Each time an object is missed, the scene goes dark for one second.

1. Propose a communication scenario between the user and the application**.....04 pts**

Start and end symbols ..... ***(0.5 pt)***
Condition symbol to distinguish between capture and loss of the object ..... ***(01 pt)***
Loop in the diagram ..... ***(01 pt)***
Correct ordering of activities ..... ***(1.5 pt)***

2. Considering the following initial plan, fill in the following tables with keywords (No sentences, ***A wrong choice cancels the cell answer***) **.....10 pts**



	Objects	Background	Wheels	Cart
Object type (Actor, Decors, Hollow Entity)	**Actor**	**Actor / Hollow Entity**	**Actor**	**Actor**
Animation nature (What geometric transformation ?)	**Translation**	**Color Interpolation**	**Rotation / Translation**	**Translation**
Animation type (Linear, NonLinear, …)	**Not Defined**	**Not Defined**	**Linear**	**Not Defined**
Animation Loop (yes, No)	**No**	**Yes**	**Yes**	**Yes**
Interactivity (User-Guided, Scenario-Defined)	**SD**	**SD**	**UG**	**UG**
Motion Speed (millimeter per second)				
Motion direction (LeftToRight, RightToLeft, CCW,CW…)	**UpToDown**	**ColorChanging**	**CCW, CW, RightToLeft, LeftToRight**	**RightToLeft, LeftToRight**
The modified attributes	**Position**	**Color**	**Position**	**Position**
The representation (Bitmap or Vector)	**Vector**	**Bitmap**	**Vector**	**BitMap**